LAUNCH COHORT · 16 AUGUST 2026 · 8:00 AM – 11:00 AM IST

Agentic AI Developer Bootcamp

Foundations, Tools, Cloud & Starter Projects for AI Engineers.
Six weeks to real fluency in the Agentic AI developer stack.

DURATION	NEXT COHORT	LIVE SESSION TIME	LEVEL
6 Weeks + 2 Pre-Bootcamp Classes	**Sun 16 Aug 2026**	**8:00 AM – 11:00 AM IST**	**Foundation to Intermediate**

- Every essential framework — LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, Agno
- Model Context Protocol (MCP) taught at working developer depth
- Cloud AI foundations — AWS Bedrock · Azure OpenAI · GCP Vertex AI
- Six weekly hands-on project artifacts + one final portfolio build
- Live cohort teaching + full recordings + learner community

PROGRAM AT A GLANCE

Six weeks to real fluency.

The Agentic AI Developer Bootcamp assembles the essential tools, frameworks, and cloud services that modern AI engineers work with into one coherent, hands-on curriculum. Every week ends with a project artifact you can point to. By the end of six weeks, you have real ecosystem fluency and a working portfolio.

DURATION	6 weeks + 2 pre-bootcamp Python foundation classes	FINAL PROJECT	Portfolio-style Agentic AI build submitted as homework
NEXT COHORT STARTS	Sunday, 16 August 2026	RECORDINGS	Every session available in your Manifold AI Learning portal
LIVE SESSION TIME	8:00 AM – 11:00 AM IST	COMMUNITY	Active learner group for questions, peer builds, and progress
LIVE SESSIONS	Every week, structured and hands-on	PREREQUISITES	Some programming exposure helps. Python covered in pre-bootcamp.
WEEKLY DELIVERABLE	One hands-on project artifact per week	DELIVERY PLATFORM	Manifold AI Learning portal

WHAT YOU WILL BE ABLE TO DO

Real engineering outcomes.

- ✓ Ship agents that call tools, structure outputs, and handle real errors — not chat wrappers.
- ✓ Wire tool schemas and function calls with judgment — expose the right surface and catch failure.
- ✓ Use Claude Code, Cursor, and GitHub Copilot as core developer tools every day.
- ✓ Build RAG assistants with LangChain and LlamaIndex — chunking, embeddings, retrievers, failure modes.
- ✓ Design graph-based agent workflows in LangGraph — state, routing, memory, human-in-the-loop.
- ✓ Compare and use multi-agent frameworks — CrewAI, AutoGen, Agno / Phidata — and know when each fits.
- ✓ Work with the Model Context Protocol (MCP) at real developer depth.
- ✓ Navigate cloud AI foundations across AWS Bedrock, Azure OpenAI, and GCP Vertex AI.
- ✓ Add tracing, evaluation, and observability to workflows with LangSmith and Langfuse.
- ✓ Ship six weekly hands-on artifacts + one final portfolio project — every one of them in your GitHub.

CURRICULUM

Every essential tool, framework, and cloud layer — in sequence.

Two pre-bootcamp Python classes get everyone ready. Six weekly modules take you through the full Agentic AI developer stack.

PRE-BOOTCAMP · FOUNDATION CLASS 1

Python for AI Developers

Setting the developer environment right before we build.

TOPICS

Python setup, VS Code and Cursor setup, virtual environments, variables, functions, lists, dictionaries, JSON handling, file handling, error handling, basic project structure.

HANDS-ON

Build a Python script that reads a file, extracts key information, and saves structured JSON.

PRE-BOOTCAMP · FOUNDATION CLASS 2

APIs, SDKs & Developer Workflow

The developer discipline behind every serious AI application.

TOPICS

HTTP basics, API calls, API keys and secrets, environment variables, .env files, JSON request/response payloads, Python SDK usage, Git and GitHub basics, Claude Code and Cursor workflow awareness.

HANDS-ON

Build a CLI application that calls an LLM API and returns structured output.

WEEK 1

Agentic AI Foundations + LLM API Mastery

The mental model, the vocabulary, and your first working assistant.

TOPICS

Agentic AI vs Generative AI, agents vs workflows, autonomy spectrum, core agent components, LLM API fundamentals, prompting design, system prompts, user prompts, structured outputs, JSON response engineering, Cursor / Claude Code / GitHub Copilot as first-class developer tools.

HANDS-ON

Build a working AI assistant using an LLM API with structured output.

DELIVERABLE → Structured Output AI Assistant

WEEK 2

Tool Calling, Function Calling & MCP Introduction

The layer that turns a chat model into a real agent.

TOPICS

Tool calling and function calling in depth, tool schema design, tool execution flow, tool error handling, common misuse risks, Model Context Protocol (MCP), MCP client / server concept, tools / resources / prompts inside MCP, where MCP fits in modern Agentic AI systems.

HANDS-ON

Build a tool-using assistant with a calculator, file-summary tool, mock search tool, and full error handling.

DELIVERABLE → Tool-Using Agent (with MCP awareness)

WEEK 3

LangChain, LlamaIndex & RAG Basics

The retrieval spine underneath every serious AI application.

TOPICS

Document loading, chunking strategy, embeddings, vector stores, retrievers, the full RAG pipeline, LangChain and LCEL, LlamaIndex, framework comparison, RAG failure modes and how to spot them early.

HANDS-ON

Build a document Q&A; assistant using LangChain or LlamaIndex.

DELIVERABLE → Working RAG Assistant

WEEK 4

LangGraph, Memory & Agent Workflows

The graph-based agent workflow model behind modern production systems.

TOPICS

Why LangGraph exists, graph-based workflows, nodes and edges, state schemas, conditional routing, short-term memory, long-term memory patterns, human-in-the-loop, agent workflow design principles.

HANDS-ON

Build a LangGraph workflow with state, routing, tool call, memory, and a conditional branch.

DELIVERABLE → LangGraph Agent Workflow

WEEK 5

Multi-Agent Frameworks + Low-Code AI Tools

When you scale from one agent to many — and when you shouldn't.

TOPICS

CrewAI, AutoGen, Agno / Phidata, deep framework comparison, role-based agents, conversation-based agents, workflow-based agents, when multi-agent systems shine, when they become overkill, low-code / no-code AI tools — Langflow, Flowise, Relevance AI-style workflows.

HANDS-ON

Build a multi-agent research workflow with researcher, writer, and reviewer agents.

AWARENESS DEMO

Recreate a similar workflow inside a low-code / no-code tool to see the trade-offs.

DELIVERABLE → Multi-Agent Research Assistant

WEEK 6

Cloud AI Foundations, Agentic Evals & Observability

The layer that separates a demo from a system you can trust to run.

TOPICS

AWS AI foundations + Amazon Bedrock, Azure AI foundations + Azure OpenAI, GCP AI foundations + Vertex AI, deployment options, FastAPI awareness, Docker basics, LangSmith, Langfuse, AgentOps awareness, agentic evals fundamentals, trace-based debugging, golden dataset basics, LLM-as-judge basics, cost / latency / reliability / safety checks.

HANDS-ON

Instrument an existing agent or RAG workflow with tracing, a simple evaluation dataset, a response quality check, a tool failure check, and cost / latency observation.

DELIVERABLE → Evaluated Agentic Workflow

TOOLS & PLATFORMS COVERED

Real breadth — grouped by domain.

Intentional coverage across the modern Agentic AI stack. Every category maps to a real engineering discipline.

Developer Toolchain	Python · Cursor · Claude Code · GitHub Copilot · Git & GitHub · VS Code
LLM APIs & SDKs	OpenAI SDK · Anthropic SDK · LLM API fundamentals · Structured outputs · Tool calling · Function calling
Agentic Frameworks	LangChain · LCEL · LangGraph · LlamaIndex · CrewAI · AutoGen · Agno / Phidata
Model Context Protocol (MCP)	MCP client / server · Tools · Resources · Prompts · Real-world integration patterns
Retrieval & RAG	Document loading · Chunking · Embeddings · Vector stores · Retrievers · Basic RAG · Agentic RAG basics
Cloud AI Foundations	AWS AI foundations + Amazon Bedrock · Azure AI foundations + Azure OpenAI · GCP AI foundations + Vertex AI
Observability & Evals	LangSmith · Langfuse · AgentOps awareness · Trace-based debugging · Golden datasets · LLM-as-judge
Low-Code / No-Code AI	Langflow · Flowise · Relevance AI-style workflows
Deployment Foundations	FastAPI · Docker · Deployment patterns across AWS, Azure, GCP

HANDS-ON EVERY WEEK

Six weekly deliverables. Every one in your GitHub.

Week 1	**Structured Output AI Assistant**	LLM-backed assistant that returns clean, structured JSON on every call.
Week 2	**Tool-Using Agent (with MCP awareness)**	Agent that calls tools, handles errors, and works inside MCP concepts.
Week 3	**Working RAG Assistant**	Document Q&A; over your own knowledge source with LangChain or LlamaIndex.

Week 4	**LangGraph Agent Workflow**	Graph-based agent with state, memory, routing, and conditional flow.
Week 5	**Multi-Agent Research Assistant**	Researcher, writer, and reviewer agents collaborating on a real task.
Week 6	**Evaluated Agentic Workflow**	One of your builds instrumented with tracing, evals, and observability.

FINAL PORTFOLIO PROJECT

A build that proves you can wire an Agentic AI system end-to-end.

The closing artifact from the bootcamp is a starter Agentic AI system you design and build yourself. Choose from four project archetypes. Every project must include the core components that reflect what you learned across the six weeks.

CHOOSE ONE ARCHETYPE

- → 1. Personal Knowledge Assistant
- → 2. Research Assistant
- → 3. Business Workflow Assistant
- → 4. Tool-Using Productivity Agent

EVERY PROJECT MUST INCLUDE

- ✓ LLM API usage
- ✓ Prompting and structured output
- ✓ At least one tool
- ✓ Basic RAG or a knowledge source
- ✓ A simple agent workflow
- ✓ Basic tracing or logging
- ✓ A simple evaluation checklist

SUBMISSION FORMAT

GitHub repository or zip file · Short reflection note (what worked, what failed, what you would improve) · Optional: demo video and architecture diagram. This is the piece you show yourself first — and then the world, when the moment is right.

WHAT MAKES THIS DIFFERENT

Depth without gatekeeping. Structure without fluff.

One structured pathway	Every essential tool, framework, and cloud foundation inside a coherent 6-week curriculum.
Framework judgment, not checklists	Know when LangChain beats LlamaIndex, LangGraph beats CrewAI, MCP beats a plain tool call.
Six weekly artifacts	No week ends without something in your GitHub. Your portfolio grows every session.
Cloud AI from day one	AWS Bedrock, Azure OpenAI, and GCP Vertex AI — first-class foundations, not afterthoughts.
Evaluation and observability built in	LangSmith, Langfuse, tracing, golden datasets, LLM-as-judge from day one.

Live teaching + community + recordings	You're not watching a course. You're inside a cohort.

NEXT STEP AFTER THIS BOOTCAMP

Real fluency now. Production mastery when you're ready.

You'll leave the Developer Bootcamp with real ecosystem fluency, six hands-on artifacts, and a portfolio-grade final build. From there, most learners keep going.

If you want to go deeper — into **production architecture, advanced RAG, evaluation depth, agent monitoring, deployment, security, system design, and enterprise-grade AI systems** — the natural next step is our advanced flagship program:

ADVANCED FLAGSHIP PROGRAM

Agentic AI Enterprise Mastery: Production Systems Bootcamp

Designed for engineers ready to architect, build, and operate real Agentic AI systems at production scale — the kind of work that ships to real users, real customers, and real stakes.

You don't need to commit today. Complete this bootcamp, ship your project, sit with what you've built — and step into Enterprise Mastery when you're ready to go deeper.

READY TO ENROL?

Six weeks. The complete Agentic AI developer stack.

Launch Cohort starts **Sunday, 16 August 2026** · Live sessions **8:00 AM – 11:00 AM IST**

Visit **manifoldailearning.com** to enrol or reach out to **support@manifoldailearning.com**

© 2026 Manifold AI Learning. All rights reserved. This program is a paid learning experience. Framework and product names are trademarks of their respective owners.